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what is transformers and explain the transformer architecture

Answer

Transformers are a type of neural network architecture that replaces recurrent or convolutional layers with self-attention mechanisms. They are designed to process every other element in parallel, which improves parallelism and captures long-range dependencies more efficiently.

Transformer architecture (as described in the provided context)

Component	Description
Overall layout	Two halves: an encoder that reads the entire input sequence, and a decoder that generates the output sequence one token at a time. Both halves use self-attention networks.
Encoder	• Stack of N = 6 identical layers . • Each layer has two sub-layers: 1. Multi-head self-attention (allows each token to attend to all other tokens in the input sequence). 2. Residual connections around each sub-layer followed by layer normalization . The encoder outputs vectors of dimension d_model = 512 .
Decoder	Mirrors the encoder structure but includes an additional masked self-attention sub-layer to prevent a token from attending to future tokens, and a cross-attention sub-layer to attend to the encoder's output.
Variations	The table lists many hyper-parameter changes (e.g., different N , d_model , d_ff , number of heads h , dropout P_drop , label smoothing ε_ls).
Key design choices	• Multi-head attention splits the attention into several heads to capture diverse relationships. • Scaled dot-product attention (not detailed in the provided context).

In short, a Transformer is a sequence-to-sequence model built from repeated blocks of self-attention and feed-forward layers, with residual connections and layer normalization. It has achieved state-of-the-art performance on tasks like machine translation.

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Figure 1: The Transformer - model architecture. The Transformer follows this overall architecture using stacked self-attention and point-wise, fully connected layers. The architecture is divided into two main parts: the encoder and the decoder. The encoder is composed of a stack of $N = 6$ identical layers. Each layer has two sub-layers: the first is a multi-head self-attention network, and the second is a simple, position-wise fully connected feed-forward network. We employ a residual connection [11] around each of the two sub-layers, followed by layer normalization. To facilitate these residual connections, all sub-layers in the model have the same output dimension $d_{model} = 512$.

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Table 3: Variations on the Transformer architecture. Unlisted values are identical to those of the base model. All metrics are on the English-to-German translation task using a wordpiece, according to our byte-pair encoding, and should not be compared to per-word perplexities. N d_{model} d_{ff} h d_k d_v P_{drop} ϵ_{ls} train PPL BLEU params step
25.8 65 (A) 1 512 512 5.29 24.9 4 128 128 5.00 25.5 16 32 32 4.91 25.8 32 16 16 5.01 25.4 (B) 16 5.16 25.1 58 32 5.01 25.4 60 (C) 2 6.11 23.7 36 4 5.19 25.3 50 8 4.88 25.5 80

25.4 53 4096 4.75 26.2 90 (D) 0.0 5.77 24.6 0.2 4.95 25.5 0.0 4.67 25.3 0.2 5.47 25.7 (E) positional embedding instead of sinusoids 4.92 25.7 big 6 1024 4096 16 0.3 300K search as described in the previous section, but no

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best models from the literature. We show that the Transformer generalizes well to other tasks by applying it successfully to English constituency parsing both with l order is random. Jakob proposed replacing RNNs with self-attention and started the effort to evaluate this idea. Ashish, with Illia, designed and implemented the fi every aspect of this work. Noam proposed scaled dot-product attention, multi-head attention and the parameter-free position representation and became the othe implemented, tuned and evaluated countless model variants in our original codebase and tensor2tensor. Llion also experimented with novel model variants, was r visualizations. Lukasz and Aidan spent countless long days designing various parts of and

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collected through web sources. This data contains private information; therefore, many LLMs employ heuristics-based methods to filter information such as names information. 2.9. Architectures Here we discuss the variants of the transformer architectures used in LLMs. The di fference arises due to the application of Figure 4: / taken from [93]. Figure 5: An example of language model training objectives, image from [93]. the attention and the connection of transformer blocks. An il- lustrati Figure 4. Encoder Decoder: This architecture processes inputs through the encoder and passes the intermediate representation to the decoder to generate the outp self-attention whereas the decoder processes the sequence one after the other with implementing

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